Fundamentals 24

Siddharth K

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Problem 24.1

Let $(X, \mathcal{T})$ be a topological space where X is finite, and consider any open cover $\mathcal{O}$ of X. Since $\mathcal{O} \subseteq \mathcal{P}(X)$ and X is finite, $\mathcal{P}(X)$ is finite so $\emptyset$ is finite. So it is its own finite subcover, so X is compact.

Problem 24.2

Call this basis $\mathcal{B}$. We show the two axioms of a basis:

1. Let $x \in \mathbb{R}$. Then $x \in [x, x+1) \in \mathcal{B}$.
2. Let $x \in A, B \in \mathcal{B}$. Then we can write $A = [a, b)$ and $B = [c, d)$ for $a, b, c, d \in \mathbb{R}$. WLOG assume $a < c$. If $b \geq d$ then $A \cap B = A \in \mathcal{B}$. Otherwise, if $b < c$ then $A \cap B = \varnothing = [0, 0) \in \mathcal{B}$, and if $b \geq c$ then $A \cap B = [a, d) \in \mathcal{B}$. So in every case, $x \in A \cap B \in \mathcal{B}$.

Problem 24.4

Let $\mathcal{O}$ be any open cover of A. Consider $\mathcal{O} \cup (X \backslash A)$. Since A is closed, $X \backslash A$ is open so $\mathcal{O}' = \mathcal{O} \cup (X \backslash A)$ is an open cover of A. It is also an open cover of X because if $x \in A$ then $x \in S \in \mathcal{O}$ for some S, and otherwise $x \in (X \backslash A) \in O'$. Since X is compact, O' has a finite subcover, F. We have that $F \backslash (X \backslash A)$ is finite and covers A, so A is compact.

Problem 24.5

Assume A is closed and x is a limit point of A. We first show that $x \in A$. For contradiction, assume $x \notin A$. Then $x \in X \backslash A$, which is open. So $X \backslash A$ is an open neighborhood of x. But since x is a limit point of A, $(X \backslash A) \cap A$ must be nonempty, which is false. So A contains all its limit points.

Now, assume A contains all its limit points. We must show that $X \backslash A$ is open. Let $x \in X \backslash A$. Then there is some $U \ni x$ where $(U \cap A) \backslash \{x\} = \varnothing$. But since $x \notin A$, it must be the case that $U \cap A = \varnothing$, so $U \subseteq X \backslash A$. So every point of $X \backslash A$ has a neighborhood contained in $X \backslash A$, meaning $X \backslash A$ is open so A is closed.

Problem 24.6

Let X be compact and $S \subseteq X$ be infinite; we must show that there exists a limit point. Suppose that a limit point doesn't exist, so that, for all $x \in X$, there is $U_x \subseteq X$ where $(U_x \cap S) \backslash \{x\} = \varnothing$. This means that each U_x meets S in at most one point (x if $x \in S$). We have that $\{U_x : x \in X\}$ is an open cover of X. By compactness, there is a finite subcover $U = \{U_{x_i} : i \in [n]\}$ where

$$X = \bigcup_{i=1}^{n} U_{x_n}.$$

Since this covers S, we have that $S \subseteq U = \bigcup_{i=1}^{n} U_{x_n}$. But each U_{x_i} contains at most one point of S, so U has at most n points. This contradicts the assumption that S is infinite, so some point of X must be a limit point of S.

Problem 24.7

Since X is compact, for any $x \in X$, we can simply pick any open neighborhood $U \ni x$ and then take $K = X$, so that $x \in U \subseteq K$ where U is an open neighborhood and K is compact.

Let X be a discrete, infinite set where each singleton $\{x\}$ is defined as open. Then singletons are compact so every point is a subset of an open neighborhood that is a subset of a compact set in X.

$\mathbb{Q}$ is not locally compact. Pick any rational $q \in \mathbb{Q}$. An open neighborhood of $q \in \mathbb{Q}$ looks like $U = (q - \varepsilon, q + \varepsilon) \cap \mathbb{Q}$ and contains infinitely many rationals. But you can choose a sequence of distinct rationals converging to an irrational in $\mathbb{R}$, which has no finite subcover. So there is no compact $K \subseteq \mathbb{Q}$ with $U \subseteq K$.

The integers are locally compact but not compact. It isn't compact because the open cover $\{\{n\} : n \in \mathbb{Z}\}$ has no finite subcover. But, since infinite sets where singletons are open are locally compact, $\mathbb{Z}$ is compact.

Problem 24.8

Part 1)

Since closed intervals are compact, we can write

$$\mathbb{R} = \bigcup_{n \in \mathbb{Z}} [n, n+1],$$

and since $\mathbb{Z}$ is countable, $\mathbb{R}$ is a countable union of compact subsets, so it is σ-compact.

Part 2)

Singletons $\{x\}$ are compact because, given any open cover $\mathcal{O} = \{U_\alpha : \alpha \in A\}$ of $\{x\}$, there must be at least one $\alpha \in A$ with $x \in U_\alpha$, so $\{U_\alpha\}$ is a finite subcover of $\mathcal{O}$. So a countable set is a countable union of compact subsets, which means it is σ-compact.

Part 3)

Let $\mathcal{B} = \{B_i : i \in \mathbb{N}\}$ be a countable basis for $(X, \mathcal{T})$ where X is locally compact. For each $i \in \mathbb{N}$, choose some $x_i \in B_i$. By local compactness, there is some open U_i and compact K_i where $x_i \in B_i \subseteq U_i \subseteq K_i$. Since $\mathcal{B}$ covers X and $B_i \subseteq K_i$ for all $i \in \mathbb{N}$, we have that

$$X = \bigcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \mathbb{N}} K_i,$$

so X is a countable union of open sets, and therefore it is σ-compact.